

Scheme 1

Stage 1 Screening and Eligibility Decision Scheme

Title and abstract eligibility for the BPS chronic pain corpus

DRAFT FOR EXPERT EVALUATION. These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes in the Python workflow with an LLM (gemini-2.5-flash); the full end-to-end run is deliberately held until after expert feedback, and the refinements proposed here are awaiting sign-off before any re-run.

At a Glance

Workflow position

Pre-extraction eligibility screen, run after search and deduplication and before Stage 2 abstract coding.

Operational mode

Deterministic rule set that emits a provisional decision, confidence, and reason. Every decision is validated by a human screener in Rayyan.

Unit of analysis

One bibliographic record (title, abstract, and publication metadata).

Provenance basis

Executable screening rules plus the OSF eligibility criteria.

Research questions

Gatekeeper for RQ1, RQ2, RQ3 (defines the analysable corpus)

Tagline

Rule-based provisional machine assist with mandatory human validation

Canonical Source Paths

- `src/protocol/decision_rules/screening_rules.md`
- `src/bps_review/screening/rules.py`
- `src/review_stages/03_screening/README.md`
- `src/protocol/osf/OSF_registration-HTBMFCPR.md`

Purpose

This scheme operationalizes title and abstract screening after search and deduplication. Its function is to decide whether a record enters the review corpus for downstream coding, using a human-validatable rule set centred on biopsychosocial language, chronic pain relevance, review design, and population eligibility.

Because everything downstream inherits this decision, the scheme is deliberately conservative: it protects recall at the boundary and pushes genuinely ambiguous records into a borderline register rather than excluding them early.

Decision Fields and Controlled Values

The scheme writes one decision bundle per record. The controlled values below are the operational vocabulary.

stage1_decision

Eligibility verdict for the record.

- **include** Meets every inclusion rule with no triggered exclusion. *Choose when:* Review design, BPS term, chronic pain, adult or mixed population, English, in window. *Not this if:* Any hard exclusion trigger present. *Boundary:* Any single hard exclusion overrides inclusion.
- **exclude** At least one hard exclusion rule fires with high confidence. *Choose when:* Clear primary study, wrong population, non-English. *Not this if:* Only a soft or ambiguous signal, which is maybe. *Boundary:* Use exclude only when the trigger is unambiguous; otherwise use maybe.
- **maybe** Borderline record retained for human adjudication (proposed reinstatement, see refinement). *Choose when:* Ambiguous review type, mixed acute and chronic, unclear age or duration. *Not this if:* A clearly satisfied or clearly failed rule. *Boundary:* maybe is the honest label for genuine uncertainty and must not be used to avoid a decision that the abstract actually supports.
- **unclear** Current executable fallback state for ambiguous cases where maybe is not emitted. *Choose when:* Record cannot be resolved from available metadata. *Not this if:* A confident include or exclude. *Boundary:* See the documented divergence: the code currently collapses maybe into unclear.

stage1_reason

Controlled reason attached to exclusions and unclear cases. See the exclusion catalogue below.

stage1_confidence

Screening or rule confidence in the decision.

- **high** Decision rests on explicit, unambiguous metadata (for example an explicit non-review publication type).
- **medium** Decision rests on strong but partly inferential signals.
- **low** Decision rests on weak or conflicting signals; always pair with a logged rationale.

stage1_screened_by

Provenance of the provisional decision. Current outputs record `codex_machine_assist` before human validation.

stage1_screening_mode

Screening mode. Current outputs record `rule_based_provisional`.

Inclusion Logic

All of the following must hold for an include.

Review design

The record is a review article or review-like evidence synthesis (systematic, meta-analysis, scoping, umbrella, narrative, realist, integrative, or expert review).

BPS lexical trigger

The title or abstract explicitly contains a biopsychosocial term: biopsychosocial, bio-psycho-social, or bio psycho social.

Chronic pain focus

The focus concerns chronic pain, persistent pain, or a named chronic pain condition.

Population

The population is adult or mixed-age rather than pediatric-only.

Window and language

The record is within the operational search window and in English.

Exclusion Catalogue and Implemented Reason Labels

Each exclusion reason below is a controlled string with an operational trigger.

no biopsychosocial term in title/abstract

No explicit biopsychosocial term is present in the title or abstract.

outside operational search window

Publication date falls before or after the operational search dates configured in config/protocol.yaml.

protocol

Protocol papers without results are excluded.

commentary/editorial/letter

Commentary, editorial, and letter publication types are excluded.

animal/non-human focus

Animal-only or non-human records are excluded unless a human focus is explicit.

pediatric-only focus

Populations restricted to under 18 are excluded.

acute pain focus

Acute-only pain records are excluded when chronicity is not also explicit.

chronic pain focus unclear

Pain is present but chronic pain relevance is insufficiently clear.

non-English record

Records not in English are excluded.

review status unclear

The record cannot be confidently identified as a review or evidence synthesis from title, abstract, or publication metadata.

Proposed Decision Order

PROPOSED REFINEMENT

Awaiting expert evaluation. Not yet applied to the pipeline.

Refinement proposed for expert evaluation. Screeners currently apply the rules as an unordered checklist, which lets two screeners exclude the same record for different reasons and depresses reason-level agreement. We propose a fixed precedence so that the first triggered rule owns the recorded reason:

1. Language. 2. Publication type (review versus protocol, commentary, editorial, letter, primary study). 3. BPS lexical trigger. 4. Chronic pain focus. 5. Population. 6. Search window. The

record is excluded at the first hard trigger; if no hard trigger fires but a soft signal remains, the record is coded maybe with the residual concern logged.

Expected effect: reason-level Cohen kappa rises because the reason is deterministic given the record, and the borderline register is used consistently rather than idiosyncratically.

Borderline Handling

Ambiguous review type, mixed acute and chronic populations, unclear age group, or unclear pain duration should not be over-excluded. Borderline cases are logged with a rationale and lower confidence.

Musculoskeletal ambiguity is intentionally carried forward. Stage 3 full-text adjudication resolves it rather than Stage 1 excluding too aggressively.

Worked Examples

Illustrative records showing the target decision and the governing rule.

A biopsychosocial approach to TMJ pain

Narrative review, adult, orofacial chronic pain

Coding: `stage1_decision` = include; `stage1_confidence` = high; `reason` = all inclusion rules satisfied.

Why: Review design, explicit BPS term, chronic orofacial pain, adult population, English, in window.

WITHDRAWN: Biopsychosocial rehabilitation for upper limb repetitive strain injuries

Systematic review flagged withdrawn

Coding: `stage1_decision` = include; `stage1_confidence` = medium; `note` = withdrawn signal carried to Stage 3 triage.

Why: Eligible on design and topic, but the withdrawn signal is preserved so Scheme 4 can escalate it rather than Stage 1 silently dropping it.

A hypothetical primary trial of exercise for acute ankle sprain in adolescents

Primary study, acute, pediatric

Coding: `stage1_decision` = exclude; `stage1_reason` = acute pain focus; `stage1_confidence` = high.

Why: Multiple hard triggers. Under the proposed decision order the reason recorded is publication type first; here shown as topic for contrast.

Documented Divergences

The prose decision rules permit a maybe state for borderline records. The current executable implementation operationalizes ambiguous cases as unclear rather than maybe. The refinement above proposes reinstating maybe as a first-class value so the borderline register is explicit in the data.

The ICD-11 pre-tag is not assigned at Stage 1; pain-family relevance (musculoskeletal or neuropathic) is resolved at Stage 2 and Stage 3. Stage 1 is uniform across both reviews and does not filter by pain family, so the musculoskeletal and neuropathic reviews are seeded from the same screened pool. It only avoids excluding pain-family-ambiguous records.

Primary Outputs Using This Scheme

- `src/review_stages/03_screening/outputs/stage1_screening.csv`
- `src/review_stages/03_screening/audit/stage1_screening_summary.csv`

- `src/review_stages/03_screening/audit/reliability_report.csv`